# Osptek Display

# SPECIFICATION FOR 7.09"EPD

Model No:

**EPD0709A02**

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## Glossary of Acronyms

| | |
|---|---|
| EPD | Electrophoretic Display (e-Paper Display) |
| EPD Panel | EPD |
| T-Con | Timing Controller |
| TFT | Thin Film Transistor |
| MCU | Microcontroller Unit |
| FPC | Flexible Printed Circuit |
| FPL | Front Plane Laminate |
| SPI | Serial Peripheral Interface |
| COG | Chip on Glass |

# 1. General Description

## 1.1 Overview

This is a 7.09” a-Si, active matrix TFT, Electronic Paper Display (EPD) panel. The panel has such resolution (282dpi) that it is able to easily display fine patterns. Due to its bi-stable nature, the EPD panel requires very little power to update and needs no power to maintain an image.

## 1.2 Feature

Si TFT active matrix Electronic Paper Display(EPD)

Three colors support: White, Black, Red, Yellow ,Blue, Green

Resolution: 1200×1600

Ultra-low power consumption

Super Wide Viewing Angle - near 180°

Extra thin & light

SPI interface

RoHS compliant

Operation temperature range: 0℃~50℃

## 1.3 General Specifications

| Item | Specification | Unit | Note |
|---|---|---|---|
| Outline Dimension | 117(H) x 157.7(V) x 0.9(T) | mm | (1) |
| Active Area | 108(H) × 144(V) | mm | |
| Driver Element | a-Si TFT active matrix | - | |
| FPL | Spectra™ 6 | - | |
| Pixel Number | 1200(H)×1600(V) | pixel | |
| Pixel Pitch | 90(H)x90(V) | um | |
| Pixel Arrangement | Vertical stripe | - | |
| Display Colors | Black/White/Red/Yellow/Blue/Green | - | |
| Surface Treatment | Anti-Glare, 3H | - | |
| COG Driver OTP layout | Follow BOE layout | | |
| EPD driving Waveform | Provided and downloaded by BOE | | |

1-3 General Specification

Note (1): Not including the FPC.

## 1.4 **Mechanical Specifications**

| Item | | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|
| Glass Size | Horizontal(H) | 116.8 | 117 | 117.2 | mm | |
| | Vertical(V) | 157.5 | 157.7 | 157.9 | mm | |
| | Thickness(T) | 0.8 | 0.9 | 1.0 | mm | (1) |

1-4 Mechanical Specification

Note (1): Not including the Masking Film. (2) Weight for reference only.

## 1.5 **FPC Specification**

| Item | Pin numbers | Pitch (mm) | Connector | Note |
|---|---|---|---|---|
| Golden Finger | 60 | 0.5 | CON60-0.5mm | |

## 2. Absolute Maximum Ratings

### 2.1 Absolute Ratings of Environment

| Item | Symbol | Value | | Unit | Note |
|---|---|---|---|---|---|
| | | Min. | Max. | | |
| Storage Temperature | $T_{ST}$ | -25 | +60 | ºC | (1),(3) |
| Storage Humidity | $H_{ST}$ | 40 | 70 | %RH | (1),(3) |
| Operating Ambient Temperature | $T_{OP}$ | 0 | +50 | ºC | (1),(2),(3) |
| Operating Ambient Humidity | $H_{OP}$ | 40 | 70 | %RH | (1),(2),(3) |

2-1 Absolute Ratings of Environment

Note (1): (a)70 %RH Max. (Ta ≦ 40 ºC), 40%RH Min. (Ta ≦ 60 ºC) where Ta is ambient temperature.

(b) No condensation and no frost in absolute ratings of Environment.

(c) Please store in original package without opening.

Note (2): The temperature of panel display surface area should be 10 ºC Min. and 40ºC Max. Refresh time depends on operating temperature.

Note (3): E Ink Material is Moisture and UV sensitive. The absolute rating operating environments describes the boundary conditions for updating the display while the absolute rating storage environment describe the boundary conditions for a display not updating. While displays are rated to perform according to specification for the warranty period at the absolute specified operating environment, the better the storage condition, the better the E Ink displays will perform. Similar to other moisture and UV sensitive components, we recommend that our displays be stored in temperature and humidity control environments, and whenever possible, under above defined Optimal Storage Condition, away from sunlight, to optimize their performance.

**2.2 Reliability Test Item**

| Item | Test Condition |
|---|---|
| Low Temperature Storage | -25℃ For 240H |
| High Temperature/Low Humidity Storage | 60℃ 35%RH For 240H |
| High Temperature/Low Humidity Operation | 50℃ / 30%RH For 240H |
| Low Temperature Operation | 0℃ For 240H |
| High Temperature/High Humidity Operation | 40 ℃ / 90%RH For 240H |
| High Temperature/High Humidity Storage | 60 ℃ / 80%RH For 240H |
| Thermal Shock | 1 Cycle:-25℃/30min → 60℃/30min, for 100 Cycles |
| ESD Gun | Air:+/-4KV，Contact:+/-2KV |
| Package Drop Test | Height:97cm<br>1Corner,3Edges&6Faces |
| Package Random VibrationTest | 1.04G, 10Hz ~ 500Hz, Direction : X,Y,Z for 30min |

2-2 Reliability Test Item

Note (1): No condensation and no frost during test. End of test, function, mechanical, and optical shall be satisfied.

Note (2): The test result and judgment are based on OTP driving waveform.

Note (3): Stay white pattern for storage and non-operation test.

## 3. Electrical Characteristics

### 3.1 Absolute Maximum Ratings of Panel

| Parameter | Symbol | Value | | Unit | Note |
|---|---|---|---|---|---|
| | | Min | Max | | |
| Supply Voltage | VDD, VDDIO | -0.5 | 3.6 | V | |
| Ground | GND | - | | - | Connect to Ground |

$T_a$ = 25 ± 2 ºC

### 3.2 Recommended Operation Conditions of Panel

| Symbol | Parameter | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| VDDIO | IO supply voltage | | 2.4 | 3.0 | 3.6 | V |
| VDD | Logic Voltage supply | | 2.4 | 3.0 | 3.6 | V |
| AVDD | Logic Voltage supply | | 2.4 | 3.0 | 3.6 | V |
| VCC | Logic Voltage supply | | 1.3 | 1.35 | 1.4 | V |
| VGH | Positive Gate driving voltage | | 26 | 27 | 28 | V |
| VGL | Negative Gate driving voltage | | -21 | -20 | -19 | V |
| VSPH | Positive source driving voltage 1 | | - | Adjusted | - | V |
| VSPH2 | Positive source driving voltage 2 | | - | Adjusted | - | V |
| VSNH | Negative source driving voltage 1 | | - | Adjusted | - | V |
| VSNH2 | Negative source driving voltage 2 | | - | Adjusted | - | V |
| $V_{IL}$ | Low level input voltage | Digital input pins | GND | - | 0.2×VDD | V |
| $V_{IH}$ | High level input voltage | Digital input pins | 0.8×VDD | - | VDD | V |
| $V_{OH}$ | High level output voltage | Digital output pins, $I_{out}$= 1 mA, VDD=2.4V | 0.8×VDD | - | - | V |
| $V_{OL}$ | Low level output voltage | Digital output pins, Iout= 1 mA, VDD=2.4V | GND | - | 0.2×VDD | V |
| $I_{STB}$ | stand-by current | Stand-by mode | - | TBD | - | uA |
| $I_{DSLP}$ | deep sleep current | Deep sleep mode | - | TBD | - | uA |

$T_a$ = 25 ± 2 ºC

Note: The Module operating current data is measured by using Oscilloscope, and extract the Mean value.

- The typical power consumption is measured using associated 25C waveform with following pattern transition: from full white pattern to color stripe pattern. (Note 3-1)

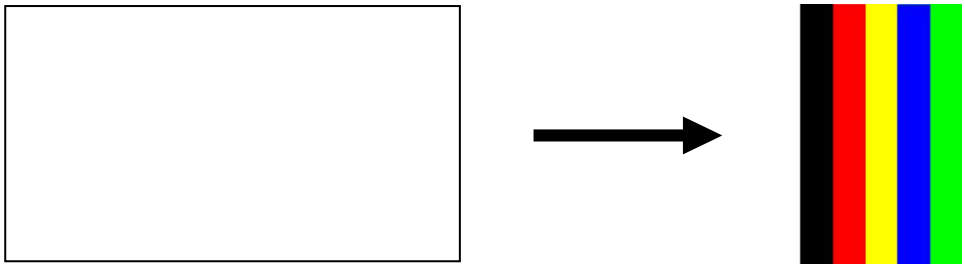
- The high loading power consumption is measured using associated 25C waveform with following pattern transition: from full white pattern to noise pattern (including random scattering of 4 colors)(Note 3-2)

- The minimum VDD value by 2.4V is based on typical application pattern with stable and continuing power supply. It does not apply on high loading pattern.

- Vcom value has been set in the IC chip on the panel.

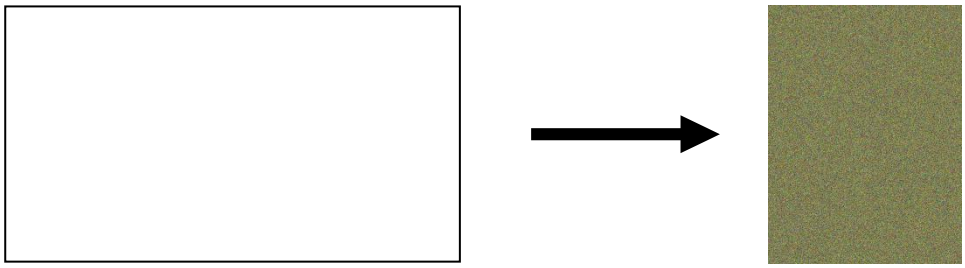
**Note 3-1**

The maximum power consumption



**Note 3-2**

The Typical power consumption



## 4. Terminal Pin Assignment & Reference Circuit

### 4.1 Terminal Pin Assignment

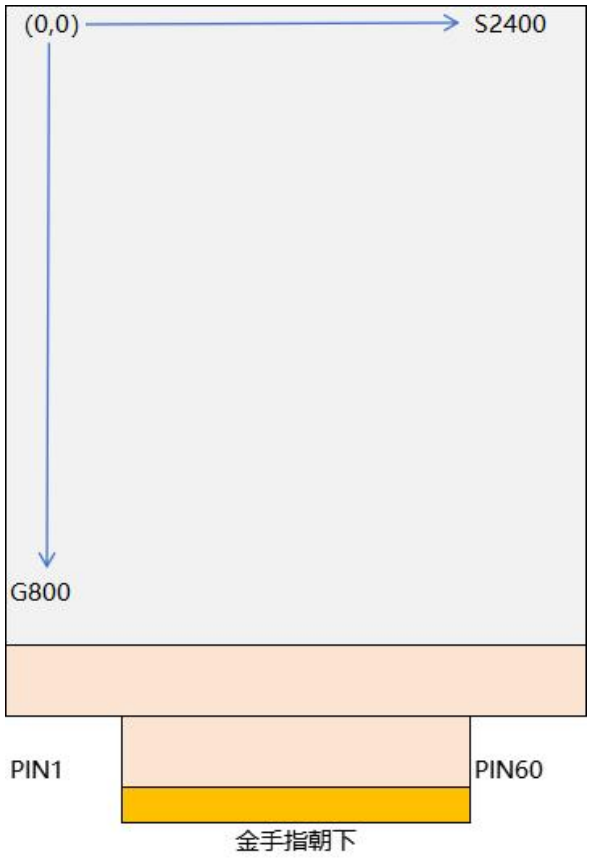
| No. | Signal | Typ | Function |
|---|---|---|---|
| 1 | VCOMBD_ | P | VCOMBD driving voltage (Master) |
| 2 | RESEC | I | Current sense input pin for the control loop.(VCOM) |
| 3 | GDRC | O | N-Channel MOSFET gate drive control.(VCOM) |
| 4 | RESEN | I | Current sense input pin for the control loop. (VDDN) |
| 5 | GDRN | O | P-Channel MOSFET gate drive control pin. (VDDN) |
| 6 | RESEP | I | Current sense input pin for the control loop. (VDDP) |
| 7 | GDRP | O | N-Channel MOSFET gate drive control pin. (VDDP) |
| 8 | GND | P | Ground |
| 9 | AVDD | P | 2.3~6V for analog voltage supply.(DC/DC) |
| 10 | VDD | P | 2.3~3.6V for analog /digital voltage supply. |
| 11 | TFT_RELAY | | |
| 12 | VCC | P | LDO output pin. |
| 13 | VDDP | P | Positive power supply for analog circuit. |
| 14 | VDDP | P | Positive power supply for analog circuit. |
| 15 | VPP | | OTP program power. |
| 16 | VDDN | P | Negative power supply for analog circuit. |
| 17 | VDDN | P | Negative power supply for analog circuit. |
| 18 | VCC1 | P | Positive power supply only for Oscilltor. |
| 19 | VDDIO | P | Power input for IO. |
| 20 | TSCL | I/O | I2C Interface to digital temperature sensor Clock pin |
| 21 | TSDA | I/O | I2C Interface to digital temperature sensor Data pin |
| 22 | BS_[0] | I | Input interface setting. Select 3 wire/ 4 wire/ Quad SPI |
| 23 | BS_[1] | I | Input interface setting. Select 3 wire/ 4 wire/ Quad SPI |
| 24 | RST_N | I | Reset. |
| 25 | BUSY_N | O | This pin indicates the driver status. Connect with a pull up |
| 26 | D_CX | I | Command/Data input. L: command H: data. (4-wire SPI). |
| 27 | CSB_M | I | Serial communication chip select.(Master) |
| 28 | SCL | I | Serial communication clock input. |
| 29 | SI0 | I/O | Serial communication data input/output (3-wire/4-wire SPI). |
| 30 | SI1 | I/O | Serial communication data input. Serial communication data |
| 31 | SI2 | I | Serial communication data input. (Standard 4-wire SPI Quad |
| 32 | SI3 | I | Serial communication data input. (Standard 4-wire SPI Quad |
| 33 | DRVP_Gate | O | Driving external N-MOSFET(VGP) |
| 34 | FBP | I | Positive charge pump(VGP) feedback pin. |
| 35 | GND | P | Ground |

| | | | |
|---|---|---|---|
| 36 | VCOMBD_S | O | VCOMBD driving voltage (Slave) |
| 37 | DRVN | O | Driving external BTJ. (VGN) |
| 38 | FBN | I | Negative charge pump feedback pin. (VGN) |
| 39 | REG_VGN | O | VGN internal reference voltage output. |
| 40 | DRVN2 | O | Driving external BTJ. (VNCP_3P5V) |
| 41 | VNCP_3P5 | P | Negative power supply for analog bias1 circuit. (for |
| 42 | VSPH | P | Positive Source driving voltage. |
| 43 | VSPL | P | Positive Source driving voltage. |
| 44 | VSPL2 | P | Positive Source driving voltage. |
| 45 | VSPHI | P | Positive source voltage. |
| 46 | VSPLI | P | Positive source voltage. |
| 47 | VSPL2I | P | Positive source voltage. |
| 48 | VSNH | P | Negative source buffer output. |
| 49 | VSNL | P | Negative source buffer output. |
| 50 | VSNL2 | P | Negative source buffer output. |
| 51 | VSNL2I | P | Negative source voltage. |
| 52 | VSNLI | P | Negative source voltage. |
| 53 | VSNHI | P | Negative source voltage. |
| 54 | FPL_VCOM | O | FPL_VCOM driving voltage |
| 55 | TFT_VCOM | O | TFT_VCOM driving voltage |
| 56 | VBB_3P5V | P | Negative power supply for analog bias2 circuit. (for |
| 57 | TFTVCOM_ | - | TFT_VCOM_SENSE |
| 58 | CSB_S | I | Serial communication chip select.(Slave) |
| 59 | VGH | P | Positive Gate driving voltage |
| 60 | VGL | P | Negative Gate driving voltage |

Note:

Type: I: Input
O: Output
C: Capacitor
P: Power

## 4.2 Panel Scan Directions



# 5. Panel AC Characteristics

## 5.1 MCU Interface Selection

In this module , there are 4-wire SPI and 3-wire SPI that can communicate with MCU.The MCU interface mode can set by hardware selection on BS1 pins. When it is “Low ”,4-wire SPI is selected. When it is “High”, 3-wire SPI(9 bits SPI) is selected.

| Pin Name | Data/Command Interface | | Control Signal | | |
|---|---|---|---|---|---|
| Bus interface | SDA | SCL | CS# | D/C# | RES# |
| SPI4 | SDIN | SCLK | CS# | D/C# | RES# |
| SPI3 | SDIN | SCLK | CS# | L | RES# |

Table 5-1 MCU interface assignment under different bus interface mode

Note: L is connected to VSS; H is connected to VCI

## 5.2 MCU Serial Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCL, serial data SDA，D/C#，CS#.

| Function | CS# | D/C# | SCL |
|---|---|---|---|
| Write Command | L | L | ↑ |
| Write data | L | H | ↑ |

Table 5-2 Control pins of 4-wire Serial Peripheral interface

Note: ↑ stands for rising edge of signal

SDA is shifted into an 8-bit shift register in the order of D7, D6, ... D0. The data byte in the shift register is written to the Graphic Display Data RAM (RAM) or command register in the same clock. Under serial mode, only write operations are allowed.

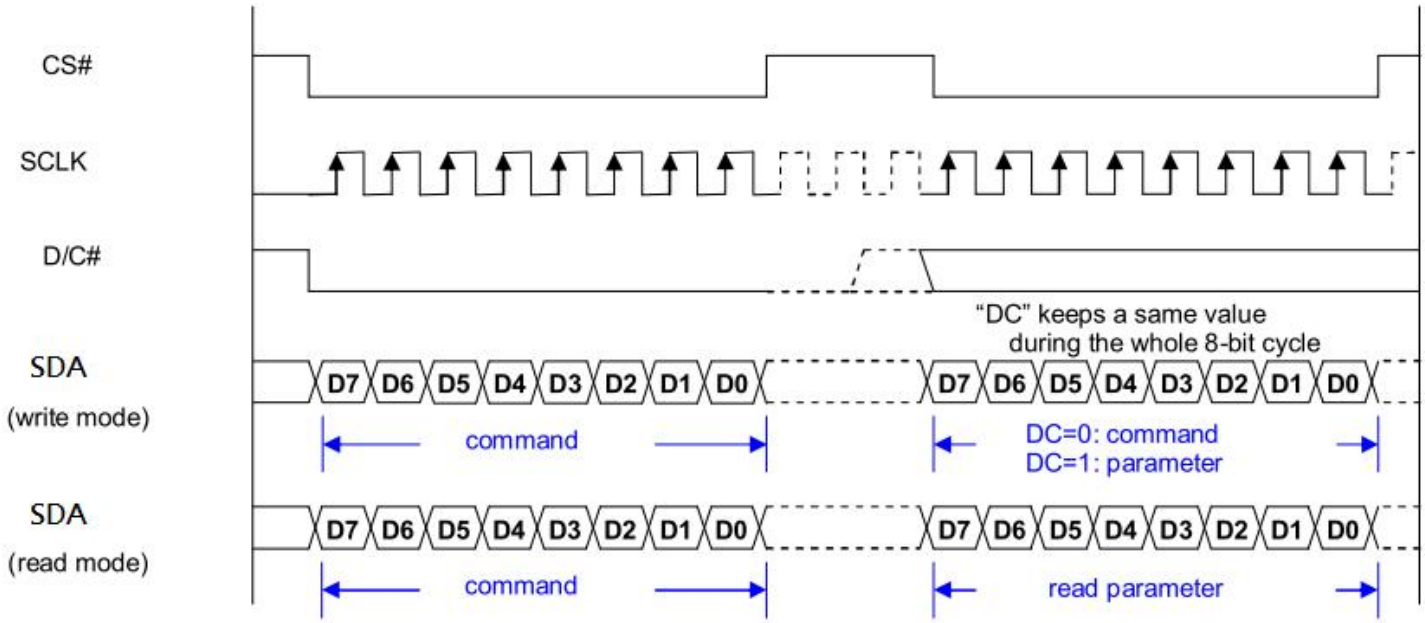



Figure 5-1 Write procedure in 4-wire SPI mode

### 5.3 MCU Serial Interface (3-wire SPI)

The 3-wire serial interface consists of serial SCL，serial data SDA and CS#.

In 3-wire SPI mode，the pin D/C# can be connected to an external ground.

The operation is similar to 4-wire serial interface while D/C# pin is not used. There are altogether 9-bits will be shifted into the shift register on every ninth clock in sequence: D/C# bit , D/C bit, D7 to D0 bit，The D/C# bit (first bit of the sequential data )will determine the following data byte in shift register is written to the Display Data RAM(D/C# bit =1 ) or the command register (D/C# bit = 0 ). Under serial mode , only write operations are allowed.

| Function | CS# | D/C# | SCL |
|---|---|---|---|
| Write command | L | Tie to Low | ↑ |
| Write data | L | Tie to Low | ↑ |

Table 5-3 Control pins of 3-wire Serial Peripheral interface

Note: ↑ stands for rising edge of signal

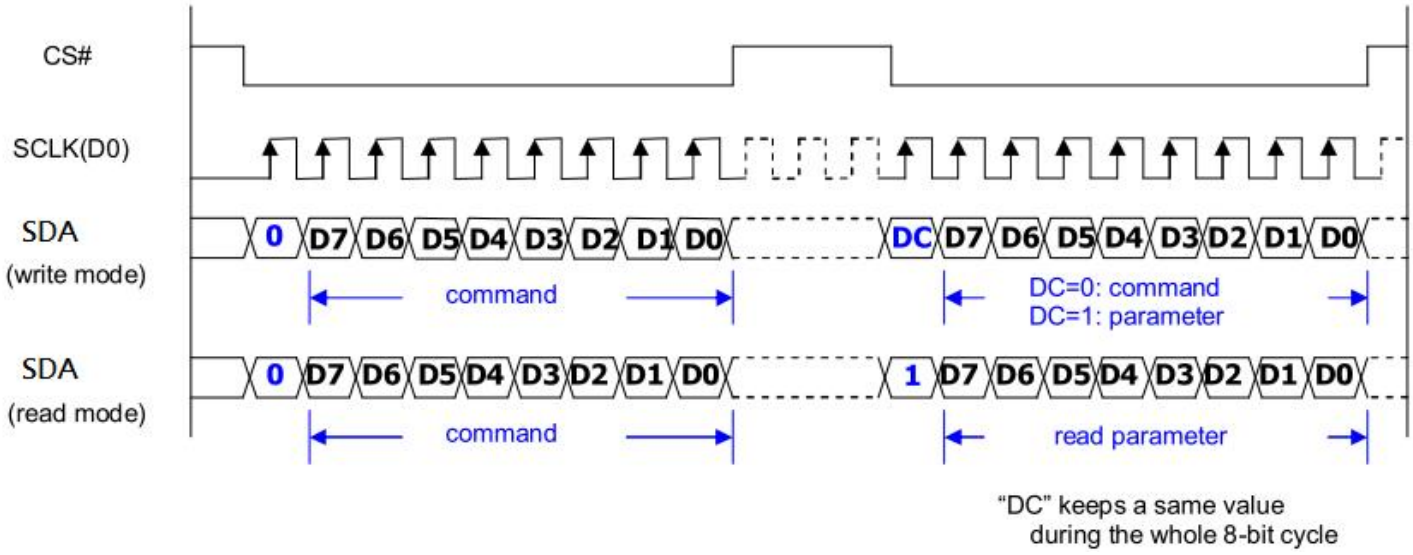



Figure 5-2 Write procedure in 3-wire SPI mode

## 6. Optical Characteristics

### 6.1 Test Conditions

Measurements are made with that the illumination is under an angle of 45 degrees, the detector is perpendicular unless otherwise specified.

| Symbol | Parameter | Conditions | Temperature | Min | Typ. | Max | Unit | Note |
|---|---|---|---|---|---|---|---|---|
| R | Reflectance | White | 25℃ | 30 | 34 | - | % | Note 6-1 |
| CR | Contrast Ratio | - | 25℃ | 15 | 22 | - | - | - |
| $T_{update}$ | Update time | - | 25℃ | - | 27 | - | sec | - |

### 6.2 Optical Specifications

| Symbol | Parameter | Conditions | Temperature | L* Typ. | a* Typ. | b* Typ. | △E2000 Max. | Note |
|---|---|---|---|---|---|---|---|---|
| WS | White State L*/a*/b* value | White | 25℃ | 66 | -4 | -2 | 6 | Note 6-1 |
| DS | Dark State L*/a*/b* value | Dark | 25℃ | 10 | 9.5 | -11 | 6 | Note 6-1 |
| RS | Red State L*/a*/b* value | Red | 25℃ | 26.5 | 41 | 29 | 6 | Note 6-1 |
| YS | Yellow State L*/a*/b* value | Yellow | 25℃ | 65 | -12 | 67 | 6 | Note 6-1 |
| BS | Blue State L*/a*/b* value | Blue | 25℃ | 32 | 6 | -40 | 6 | Note 6-1 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| GS | Green State<br>L*/a*/b* value | Green | 25℃ | 34 | -22 | 8 | 8 | Note 6-1 |

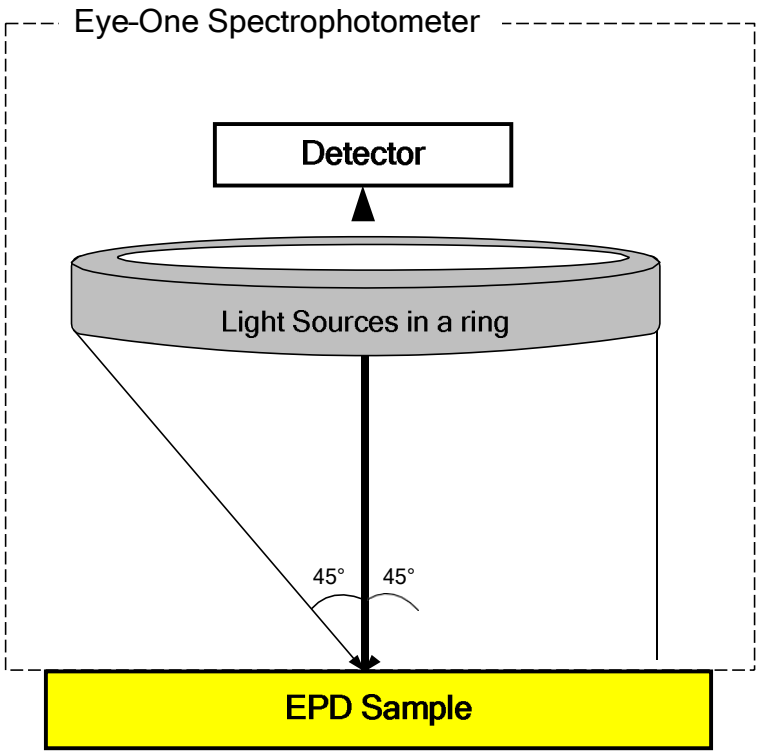
Note WS: White state, DS: Dark state, RS: Red state, YS: Yellow state, BS: Blue state, GS: Green state ;

Note 6-1: Luminance meter: Eye-One Pro3 plus Spectrophotometer

Note 6-2 ：ΔE2000 follows official CIEDE2000 color difference formula definition. The color of blue and green state will be in the range of BS and GS_ΔE2000 maximum value and the reference point is the typical value.

### 6.3 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd): CR = Rl/Rd




### 6.4 Reflection Ratio

The reflection ratio is expressed as:

R = Reflectance Factor$_{\text{white board}}$ x ( Lcenter / Lwhite board )

$L_{center}$ is the luminance measured at center in a white area (R=G=B=1). $L_{white\ board}$ is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

## 7. Packing

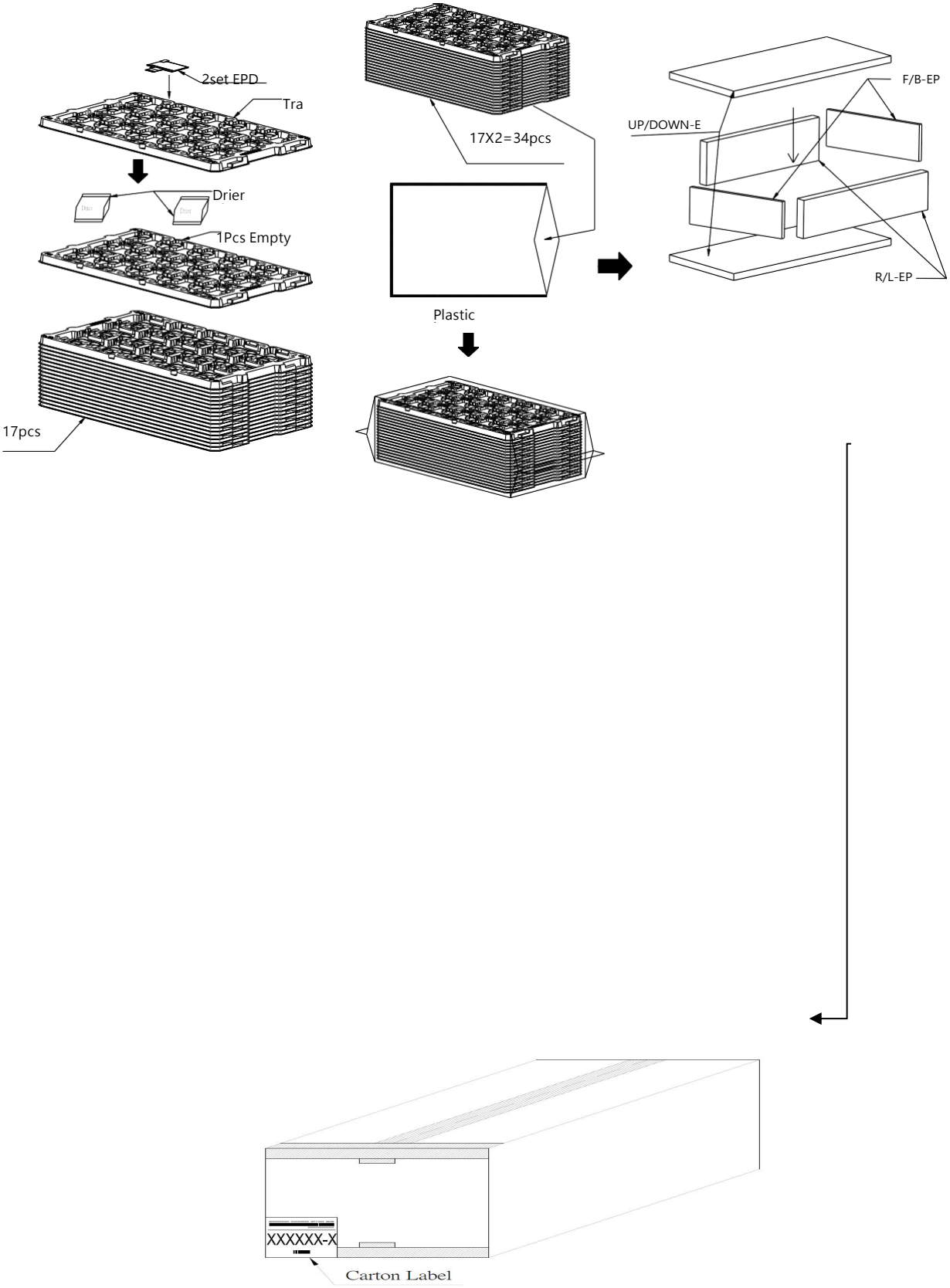
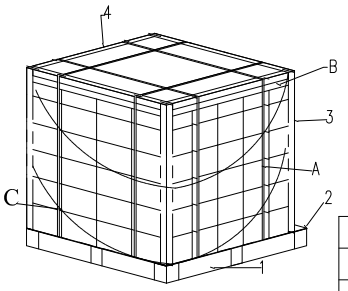



Figure 7.1 Packing

Note: The package method of Figure 7.1 is just for reference, the actual packing number refers to actual productions.




The process of pallet packing:
Put Paper Board on pallet
Stock products on pallet
Wrap the PE film
Put corner paper on corner
Tie
Wrap the PE film

| | |
|---|---|
| 1 | Pallet |
| 2 | Paper Board |
| 3/4 | Corner Paper |
| A | Tie |
| B | PE Film |
| C | PE Film |

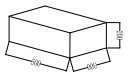
TOLERANCE TABLE(±) UNLESS OTHERWISE SPECIFIED

| DIMENSION UNITmm | Grade 1 UNITmm | Grade 2 UNITmm | Grade 3 UNITmm | Grade 4 UNITmm |
|---|---|---|---|---|
| ≤10 | ±0.05 | ±0.1 | ±0.2 | ±0.3 |
| 10-50 | ±0.1 | ±0.25 | ±0.3 | ±0.45 |
| 50-100 | ±0.15 | ±0.2 | ±0.4 | ±0.6 |
| 100-200 | ±0.2 | ±0.25 | ±0.5 | ±0.8 |
| 200-400 | ±0.25 | ±0.3 | ±0.6 | ±1.2 |
| 400-600 | ±0.3 | ±0.4 | ±0.8 | ±1.5 |
| 600-800 | ±0.4 | ±0.5 | ±1.0 | ±2.0 |
| 800-1200 | ±0.5 | ±0.6 | ±1.5 | ±2.5 |
| >1200 | ±0.8 | ±1.0 | ±2.0 | ±3.5 |
| Angle | ±0.5° | \ | ±2° | ±3° |

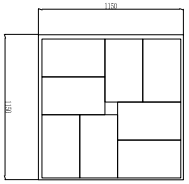
# 模组 7.09' container loading_pallet




Tray装箱时：共19个Tray,最上和最下为空Tray
Total:1*2*(19-2)=34

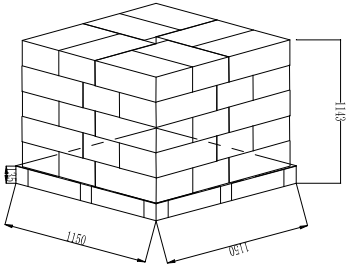


1carton=34u

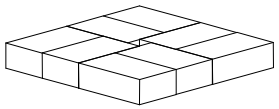



1layer=8u

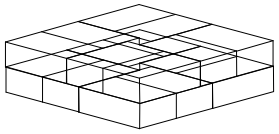
俯视图




8u/layer*5layer/p=40u/p
Total:34*40=1360PCS



一层



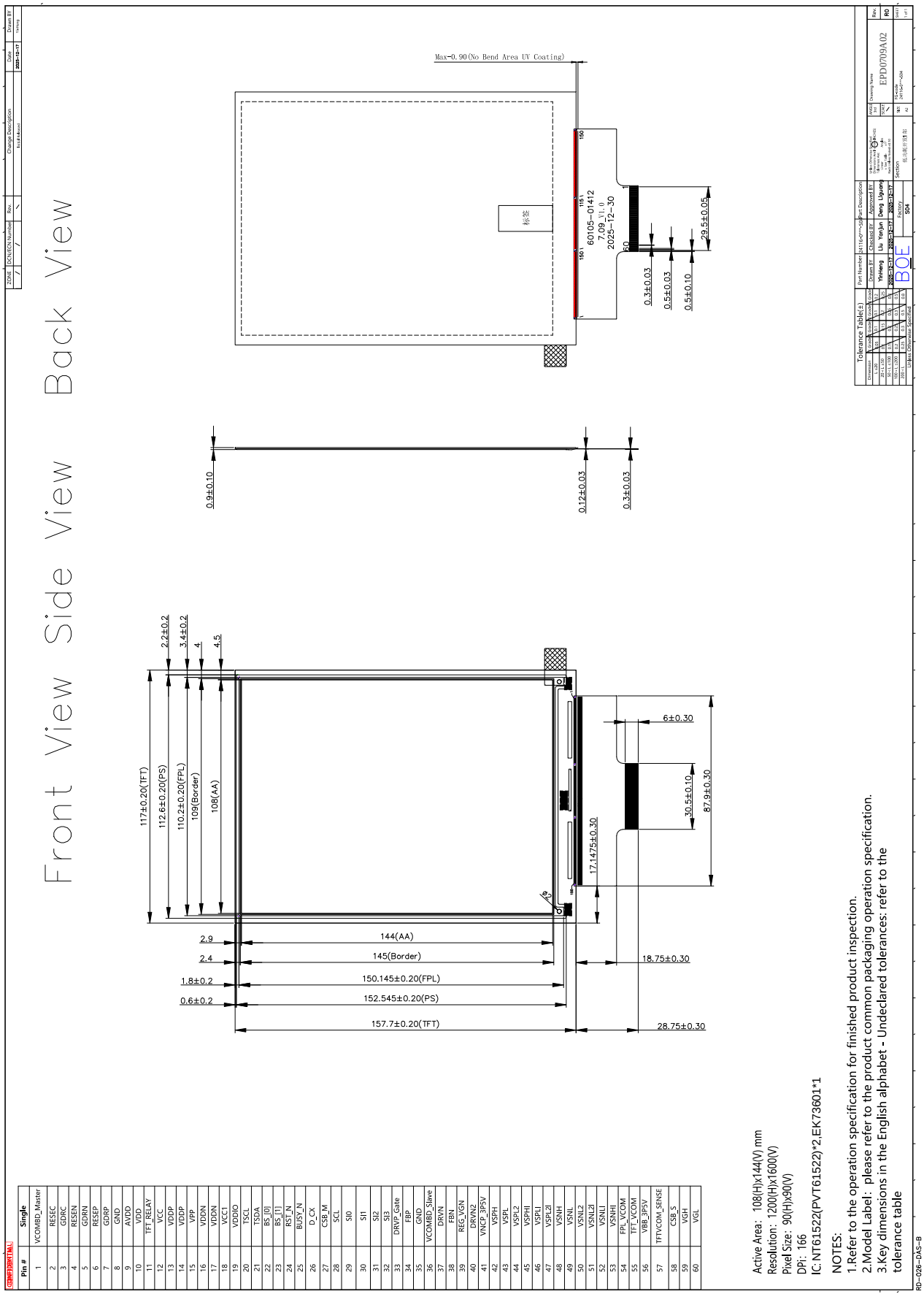
二层

| 3/4 | Paper Board | 350740000073000 | CORNER PROTECTOR_1065_50_50_3.G_纸板包材.角板 | 0.0007354 |
|---|---|---|---|---|
| \ | Corner Paper | 3011000003316000 | GIFT BOX_A+B=8.G_纸板包材.纸地板_1120*1100 | 0.000092 |
| \ | Pallet | 350710000531000 | PALLET_1150mm_1150mm_135mm_木头_烟熏 | 0.000092 |
| 序号 | 名称 | 料号 | 描述 | 数量(PCS) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | PROJECTION: | THIRD ANGLE |
| | | | DESIGNED: XYG | DATE: 2024.01.29 | UNITS: MM | PART NAME: | 7.09模组 |
| | | | DRAWN: XYG | DATE: 2024.1.29 | FINISH: NA | | |
| | | | CHECKED: | DATE: | MASS(Kg): | PART NO.: | XXX |
| \ | 2024.1.29 | FIRST ISSUE | XYG | APPROVED: | DATE: | SCALE: \ | REV: \ |
| REV | DATE | DESCRIPTION | DESIGNER | MODEL NAME: 2.13 DDM | | SHEET: \ \ \ | MATERIAL: XXX |

Figure 7.2 Stack

## 8. EPD Drawing




| Pin # | Single |
|---|---|
| 1 | VCOMBD_Master |
| 2 | RESIC |
| 3 | GDRC |
| 4 | RESEN |
| 5 | GDRN |
| 6 | RESEP |
| 7 | GDRP |
| 8 | GND |
| 9 | AVDD |
| 10 | VDD |
| 11 | TFT_RELAY |
| 12 | VCC |
| 13 | VDDP |
| 14 | VDDP |
| 15 | VPP |
| 16 | VDDN |
| 17 | VDDN |
| 18 | VCC1 |
| 19 | VDDIO |
| 20 | TSCL |
| 21 | TSDA |
| 22 | BS_[0] |
| 23 | BS_[1] |
| 24 | RST_N |
| 25 | BUSY_N |
| 26 | D_CK |
| 27 | CSB_M |
| 28 | SCL |
| 29 | SI0 |
| 30 | SI1 |
| 31 | SI2 |
| 32 | SI3 |
| 33 | DRVP_Gate |
| 34 | FBP |
| 35 | GND |
| 36 | VCOMBD_Slave |
| 37 | DRVN |
| 38 | FBN |
| 39 | REG_VGN |
| 40 | DRVN2 |
| 41 | VNCP_3P5V |
| 42 | VSPH |
| 43 | VSPL |
| 44 | VSPL2 |
| 45 | VSPHI |
| 46 | VSPLI |
| 47 | VSPLI2I |
| 48 | VSNHI |
| 49 | VSNLI |
| 50 | VSNLI2I |
| 51 | VSNLI2 |
| 52 | VSNLI |
| 53 | VSNHI |
| 54 | FPL_VCOM |
| 55 | TFT_VCOM |
| 56 | VBB_3P5V |
| 57 | TFTVCOM_SENSE |
| 58 | CSB_S |
| 59 | VGH |
| 60 | VGL |

Figure 8.1 EPD Drawing

## 9. Precaution

(1) The EPD Panel / Module is manufactured from fragile materials such as glass and plastic, and may be broken or cracked if dropped. Please handle with care. Do not apply force such as bending or twisting to the EPD panel.

(2) It is recommended to assemble or install EPD panels in a clean working area. Dust and oil may cause electrical shorts or degrade / scratch / dent the protection sheet film.

(3) Do not apply pressure to the EPD panel in order to prevent damaging it.

(4) Do not connect or disconnect the interface connector while the EPD panel is in operation.

(5) Please support the bezel with your finger while connecting the interface cable such as the FPC.

(6) Do not stack the EPD panels / Modules.

(7) Do not press the FPC on the glass edge or Pull FPC up / down to 90°.

(8) Do not touch the FPC lead connector.

(9) Do not touch IC bonding area. It may scratch TFT lead or damage IC function.

(10) Wear a Wrist Strap (Grounding connect) when handling and during assembly. Semiconductor devices are included in the EPD Panel / Module and they should be handled with care to prevent any electrostatic discharge (ESD). (An Ion Fan may be needed in assembly operation to reduce ESD risk.)

(11) Keep the EPD Panel / Module in the specified environment and original packing boxes when storage in order to avoid scratching and keep original performance.

(12) Do not disassemble or reassemble the EPD panel.

(13) Use a soft dry cloth without chemicals for cleaning. Please don't press hard for cleaning because the surface of the protection sheet film is very soft and without hard coating. This behavior would make dent or scratch on protection sheet.

(14) Please be mindful of moisture to avoid its penetration into the EPD panel, which may cause damage during operation.

(15) It's low temperature operation product. Please be mindful the temperature different to make frost or dew on the surface of EPD panel. Moisture may penetrate into the EPD panel because of frost or dew on surface of EPD panel, and makes EPD panel damage.

(16) High temperature, high humidity, sunlight or fluorescent light may degrade the EPD panel's performance. Please do not expose the unprotected EPD panel to high temperature, high humidity, sunlight, or fluorescent for long periods of time. Please store the EPD panel in controllable environment of warehouse and original package: Without sunlight, without condensation, a temperature range of 5ºC to 35ºC, and humidity from 45%RH to 70%RH.

(17) If the EPD Panel / Module is not refreshed every 24 hours, a phenomena known as "Ghosting" or "Image Sticking" may occur. It is recommended that customer refreshed the ESL / EPD Tag every 24 hours in use case with white image. It is recommended that customer ships or stores the ESL / EPD Tag with a completely white image to avoid this

issue.

(18) The label ink used for marking the Panel ID number is erased easily by solvent. Please avoid using solvent to clean the EPD panel.

(19) The EPD is vacuum packed.

(20) Before approved by BOE and customer, products and product specifications may be subject to change without notice. Confirm that you have received the latest product standards or specifications before final design, purchase or use.

(21) BOE makes every attempt to ensure that its products are of high quality and reliability. However, contact BOE sales office before using the product in an application that demands especially high quality and reliability or where its failure or malfunction may directly threaten human life or cause risk of bodily injury, such as aerospace, aeronautics, nuclear power, combustion control, transportation, traffic, safety equipment or medical equipment for life support.

(22) Design your application so that the product is used within the ranges guaranteed by BOE particularly for maximum rating, operating supply voltage range, heat radiation characteristics, installation conditions and other characteristics. BOE bears no responsibility for failure or damage when used beyond the guaranteed ranges. Even within the guaranteed ranges, consider normally foreseeable failure rates or failure modes in semiconductor devices and employ systemic measures such as fail safes, so that the equipment incorporating BOE product does not cause bodily injury, fire or other consequential damage due to operation of the BOE product.

(23) This product is not designed to be radiation resist

## 10. Definition of Labels

### 10.1 Definition of Production Label

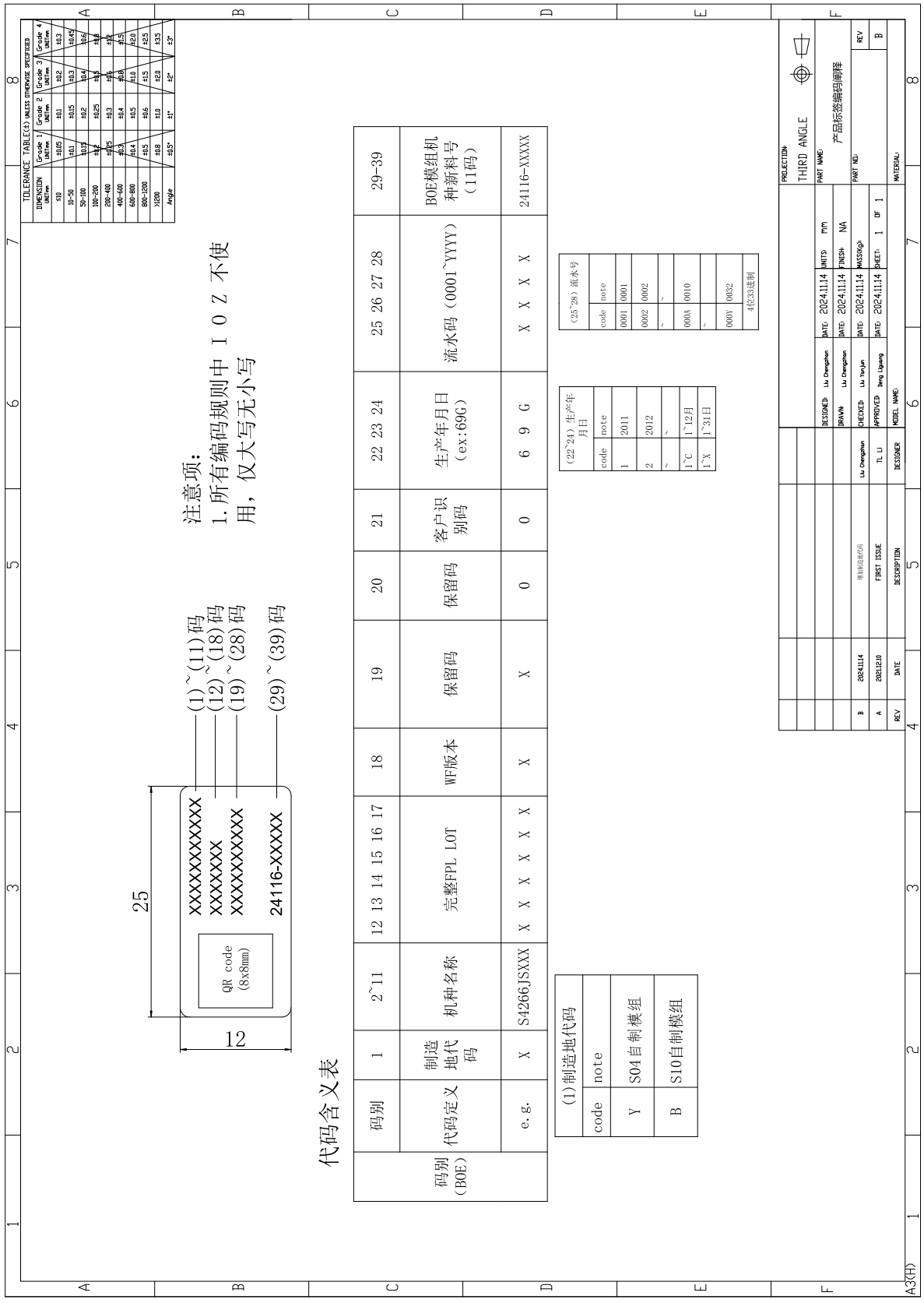



Figure 10.1 Definition of Production Label

## 10.2 Definition of Pallet Label

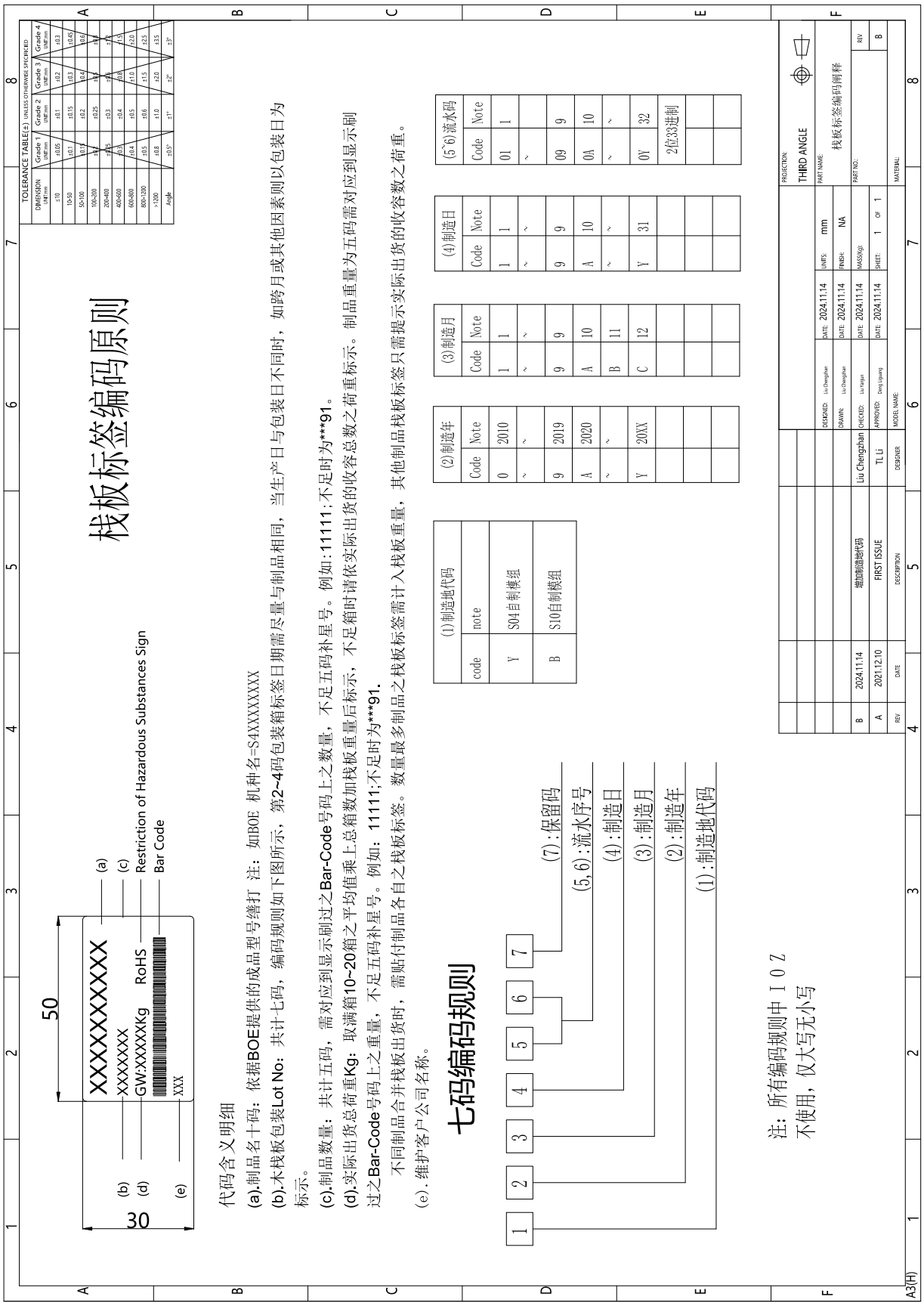



Figure 10.2 Definition of Pallet Label

### 10.3 Definition of Carton Label

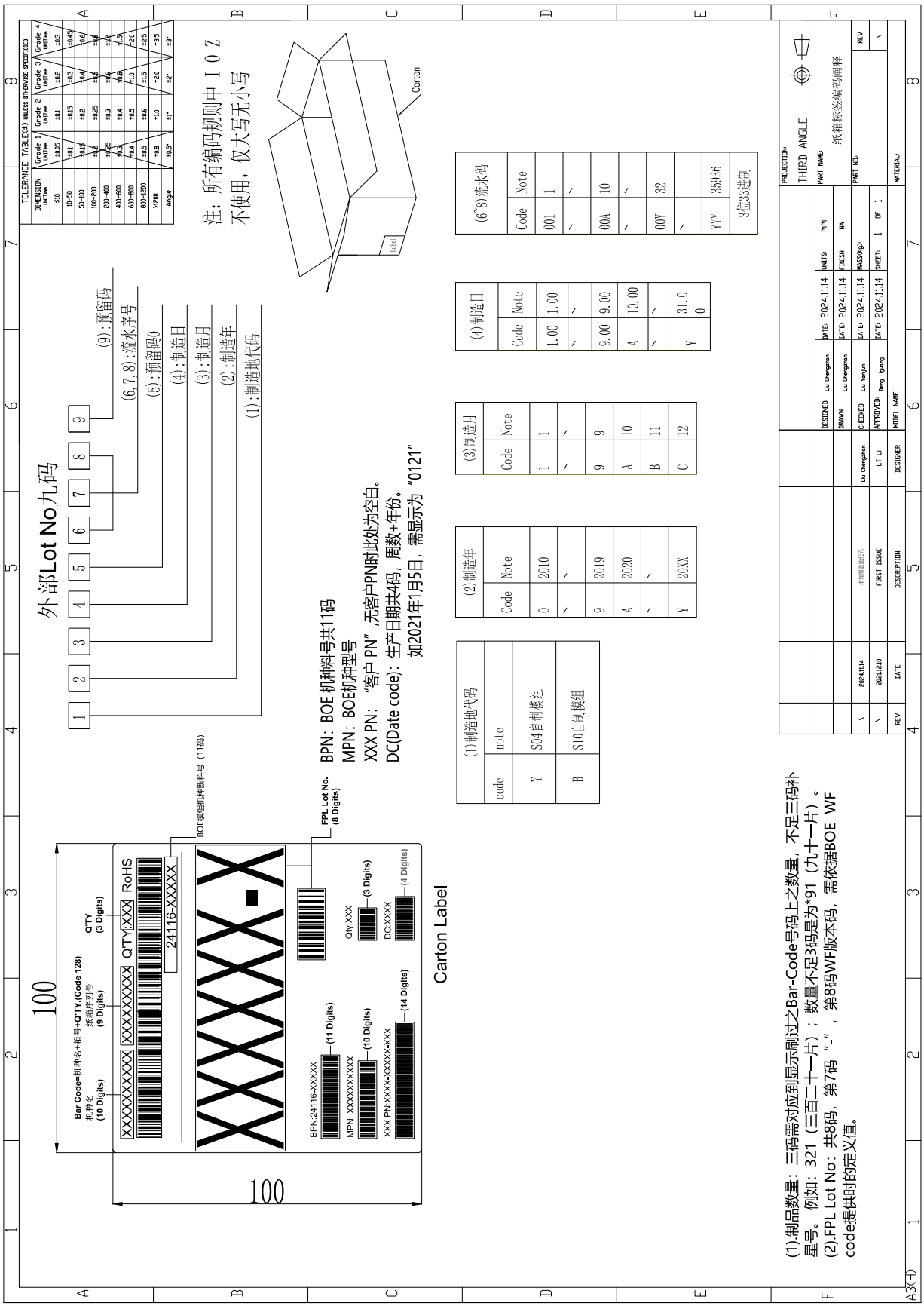



Figure 10.3 Definition of Carton Label